E34: a cheaper order-24 class representative, and the matched rho on E18′s $j = 0$ curves across the §5a band

Run report and results, 2026-11-01. Note §9.38 (pre-registration, `45378560d`) and §9.39 (results). Code and binary `bd0118f39` (new module `src/cryptanalysis/twist24_rep.rs`; E29′s functions untouched), tables `examples/e34_tables.rs`. Records `experiments/38_twist24_cheap_rep/`. Host: the shared 4-vCPU container, one core, serial; isolab-eks unreachable (no `aws`, no `kubectl`); no isolation tier; **operation counts only**, no wall-time claim. Under one CPU-minute of runs.

Construction

On the canonical model $y^2 = x^3 + b_0 t^3$, candidate (j, k) of a class is $\iota^k \circ \Phi^j(P)$. On abscissae both maps are coordinate-wise: $x_i \mapsto c_{ijk} x_i$ with $c_{ijk} = h^{2j} z^{ij} \omega^k \in \mathbb{F}_p$ ($h^2 = z^{-1}$). The cheap representative therefore (1) computes coordinate 0 of the twelve abscissae (10 products, 1 negation) and keeps the minimum with 11 comparisons, going on to the next coordinate only on a tie ($x_0 = 0$); (2) completes only the winner's abscissa and computes only the winner's ordinate (Φ^j on y is a monomial map); (3) fixes the sign by a half-p test on the first non-zero ordinate coordinate. The multiplier comes from a 24-entry table. Arms: order 6 (E29), order 6 cheap, order 24 (E29), order 24 cheap, order 24 cheap mul-lean (sign classes: 5 products and 6 negations on coordinate 0).

Checks

- **E29 replay**: counted copies reproduce all 512 E29 Stage R records in every field.
- **Representative identity**: 0 mismatches over 39,936 order-24 class members (both cheap variants) and 9,984 order-6 members (200 random points of $\langle G \rangle$ plus 8 forced $x_0 = 0$ points per curve), rep $= [m]P$ on every scalar check. No curve has a point with $x = 0$.
- **Walk identity**: 768 of 768 cheap walks equal their built walk (steps, group operations, escapes, log, correctness, visit digest).
- **Negative controls**: omitted candidate 3,328 of 39,936 mismatches and digest differs on 16 of 16 walks; wrong ordinate 39,936 of 39,936 mismatches.
- **Unit tests**: 3 of 3 pass (`cargo test --release --lib twist24_rep`). Clippy and rustfmt pass on E34′s files.

Measured (per step, schoolbook, $p \geq 541$)

Representative $\mathbb{F}_p$ products per call: E29 order 24 86–92, cheap 15.17, mul-lean 10.17, E29 order 6 8.00, cheap order 6 4.00. Multiplications per step: cheap order 24 117.2–120.2 against 109.1–111.2 (E29 order 6). Native operations per step: 25.8 (cheap order 24; 13.95 comparisons), 16.8 (E29 order 6), 65.3 (E29 order 24, of which 40.6 mod-n).

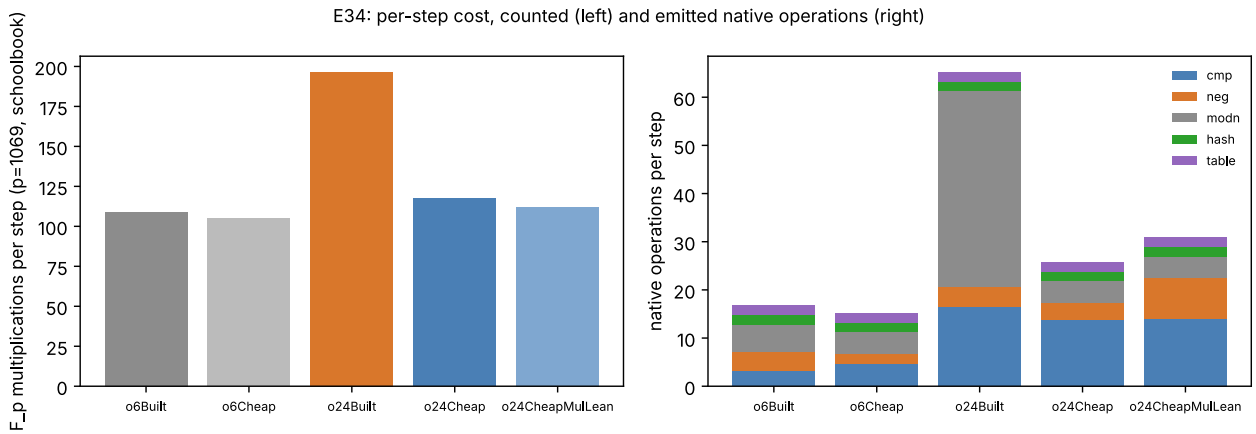


Figure 1: Per-step $\mathbb{F}_p$ multiplications (left) and emitted native operations by kind (right), $p = 1069$, schoolbook, 16 walks per arm. Source: `summary.json`.

The band (pooled order 6 / order 24; every emitted operation priced at $w \cdot 97$)

pair	arith.	0	0.1	1 GAE	holds
E29 / E29	school	0.962	0.562	0.453	yes, order 6
E29 / cheap	school	1.563	1.254	1.126	yes, order 24 (7/8 curves)
E29 / cheap	Karatsuba	1.544	1.232	1.122	yes, order 24 (7/8)

cheap / cheap	school	1.505	1.167	1.026	yes, order 24 (3/8; 5 unresolved)
cheap / cheap	Karatsuba	1.476	1.140	1.021	yes, order 24 (3/8; 5 unresolved)

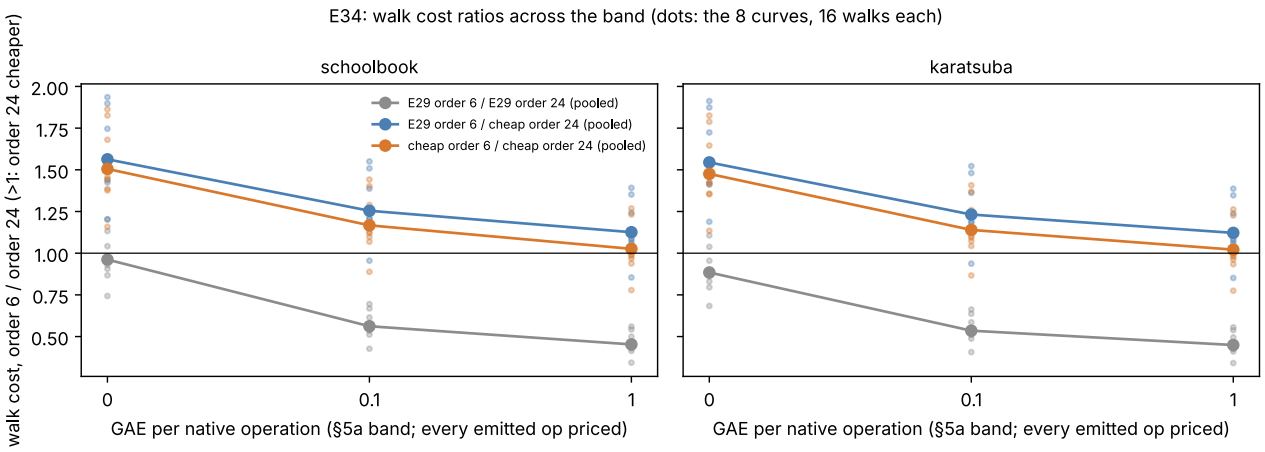


Figure 2: Walk cost ratios across the §5a band; large dots pooled, small dots the eight curves (16 walks each). Source: `summary.json`.

Verdicts (§9.38)

G2.134–G2.142 hold (G2.135′s registered count 256 was per arithmetic; 512 were compared). G2.143 (derived): E18′s ψ-arm distances lengthen by 1.632 ×, 1.254 ×, 1.126 × at 0, 0.1, 1 GAE (schoolbook; Karatsuba 1.627, 1.232, 1.122), and the modelled handover moves later by 2.83, 1.31, 0.68 bits.

What it says

With its representative built from the coordinate-wise structure, the order-24 walk is the stronger rho on E18′s $j = 0$ curves at every weight of the band, pooled, in both arithmetics, against both order-6 walks. The margin over the cheap order-6 walk at 1 GAE is 2–3% and is not resolved per curve on five of eight curves (16 walks each). With every operation emitted, E29′s own order-24 walk loses at every weight; §9.33′s crossing came from pricing one operation per step. **Class: a reference correction**: E18′s rows move further from rho; no index-calculus cost changed, and this is not an improvement. No leaderboard or `benchmarks/` row.